EAB 770 - Chapter 12

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1 Poisson Regression

1.1 Outline

- Poisson Regression
  - When is it used for?
  - Offset in Incidence Densities
  - Overdispersion
  - Exact Methods
- Generalized Linear Models

1.2 Poisson Regression

We have discussed how to model different categorical responses - Mantel-Haenszel: associations between ordinal and nominal variables - Logistic Regression: dichotomous and polytomous responses

> **Note**
>
> Counts are also a discrete variable. Typically assumed to be a normal distribution, but its discreteness means it's not a perfect fit to a continuous distribution like the normal distribution. We use the Poisson distribution to model counts.

1.3 Poisson Regression: Applications

- Examples of counts include:
  - Colony counts in bacteria
  - Accidents

  - Equipment failures
  - Rare Disease Incidence

1.4 Generalized Linear Models

- Generalized linear models (GLM) allows us to have non-Gaussian error distributions for a response variable.
  - Typically used when the response variable is **NOT** Gaussian.
- GLMs utilize a link function (η) to transform the response variable based on its assumed distribution.
  - Proportions: Logit link functions (Logistic regression)
  - Poisson: Logarithmic function

1.5 Poisson Regression: GLM

Suppose a response variable Y is distributed as Poisson and has expected value for a unit of time/space. We can write a Poisson regression model as a loglinear model such that:

$$\eta = \log \mu = \alpha + \sum_i \beta_i x_i$$

> **Note**
>
> The link function η is logarithmic in nature, which means a 1-unit increase in x_i leads to an increase of e^{β_i} in the response variable.

1.6 Offset

Sometimes, counts are measured for an interval of time (seconds, hours, weeks) or space (inches, kilometers, miles), which we designate as N.

> **Poisson Rates**
>
> The ratio Y/N can be interpreted as the rate of counts per unit time. Examples include:
>
> - Monthly mortality rate
> - Yearly failure rate
> - Daily hospitalization rate

1.7 Offset: GLM

The expected values of Y/N is μ/N. The loglinear model can now be written as:

Note

$$\log \mu/N = \alpha + \sum_i \beta_i x_i \rightarrow log\mu = \log N + \alpha + \sum_i \beta_i x_i$$

$\log N$ is called the offset term.

Note

The offset can be a function of the explanatory variables, such that $N = N(x_i)$.

Incidence Rate

We define an incidence rate $\lambda(x_i)$ is defined as:

$$\lambda(x_i) = \frac{\mu(x_i)}{N(x_i)}$$

We can rewrite the model as:

$$\eta_i = \log \lambda(x_i) = \alpha + \sum_i \beta_i x_i$$

1.8 Poisson Model

We can specify the model using the following syntax:

$$Y_i \sim Poisson(\lambda_i)$$

$$\eta_i = \log \lambda(x_i) = \alpha + \sum_i \beta_i x_i$$

1.9 Comparing Incidence Rates

Note that the linear predictor is in a logarithmic scale with respect to the incidence rates. Suppose we want to compare $ _1 = (x_1)$ and $ _2 = (x_2)$ in terms of the linear predictors η_1 and η_2

$$\eta_1 - \eta_2 = \log \lambda_1 - \log \lambda_2 = \log(\lambda_1/\lambda_2)$$

> **Note**
>
> The estimated differences in the linear predictor estimates the logarithm of the ratio of the two incidence rates.

$$\exp(\eta_1 - \eta_2) = \lambda_1/\lambda_2$$

> **Note**
>
> Exponentiating the differences in the linear predictors will provide an estimate of the incidence rate ratio, similar to the incidence density ratio discussed in Chapter 2.

1.10 Poisson Regression: `SAS`

The `GENMOD`, `GLM`, and `GLIMMIX` procedures can perform Poisson regression in SAS.

> **GENMOD procedure**
>
> The `GENMOD` procedure has very similar syntax to the `LOGISTIC` procedure.
>
> - Since `GENMOD` can handle other distributions, we need to specify the `DIST` option in the `MODEL` statement as `DIST=POISSON`.
> - Joint tests can also be displayed by adding the `TYPE3` option to the `MODEL` statement.
> - Interaction plots can be created using the `EFFECTPLOT` statement.
> - The estimated means can be calculated using the `LSMEANS` statement.
>
> The linear predictor differences and incidence rate ratios can be tested using `CONTRAST` statements OR estimated using the `ESTIMATE` statement.

> **Warning**
>
> The criteria for assessing goodness of fit in Poisson regression is the Pearson χ^2/DF, where DF is the degrees of freedom. This value is included in the goodness of fit table in the "Scaled Pearson X2" row. The Value/DF should be close to 1.

1.11 Poisson Regression: `R`

The `glm()` function can be used to estimate a Poisson GLM. The `family` should be specified as `family="poisson"`.

> **Tip**
>
> `emmeans`, `contrast` and `Anova` also works with Poisson GLMs in a similar way as the logistic regression models created using `glm()`.

> **Warning**
>
> The criteria for assessing goodness of fit in Poisson regression is the Pearson χ^2/DF, where DF is the degrees of freedom. This value is included in the output of the `check_overdispersion()` function in the `performance` package in `R` as the `dispersion ratio`.

1.12 Example

The table shows salmonella counts for samples taken from two science labs. The researchers want to determine whether the counts are different between the labs. Estimate the incidence rate ratio of salmonella between the two labs.

1.12.1 Table

Table 12.1 Salmonella Counts in Two Science Labs

Science Lab	Counts
A	63 64 65 68 69 70 72 73 75 80 82 83 83 84 84 85 90 91
B	168 171 174 175 185 189 190 191 195 197 198 198 203 205 205 207 210 214 216 218

1.12.2 SAS Data Set

```
data salmonella;
input lab $ counts @@;
datalines;
A 63 A 64 A 65 A 68 A 69 A 70 A 72 A 73
A 75 A 80 A 82 A 83 A 83 A 84 A 84 A 85 A 90 A 91
B 168 B 171 B 174 B 175 B 185 B 189 B 190
B 191 B 195 B 197 B 198 B 198 B 203 B 205 B 205
B 207 B 210 B 214 B 216 B 218
;
```

1.12.3 R Data Set

```
library(tidyverse)
```

```
-- Attaching core tidyverse packages ------------------------ tidyverse 2.0.0 --
v dplyr     1.2.1     v readr     2.2.0
v forcats   1.0.1     v stringr   1.6.0
v ggplot2   4.0.3     v tibble    3.3.1
v lubridate 1.9.5     v tidyr     1.3.2
v purrr     1.2.2
-- Conflicts ------------------------------------------ tidyverse_conflicts() --
x dplyr::filter() masks stats::filter()
x dplyr::lag()    masks stats::lag()
i Use the conflicted package (<http://conflicted.r-lib.org/>) to force all conflicts to beco
```

```
salmonella <- read.csv("salmonella.csv")
glimpse(salmonella)
```

```
Rows: 38
Columns: 2
$ lab    <chr> "A", "A", "A", "A", "A", "A", "A", "A", "A", "A", "A", "A", "A"~
$ counts <int> 63, 64, 65, 68, 69, 70, 72, 73, 75, 80, 82, 83, 83, 84, 84, 85,~
```

1.12.4 SAS Code

```
proc genmod data=salmonella;
class lab;
model counts = lab / dist=poisson link=log type3  ;
```

```
lsmeans lab/diff=all ilink cl;
estimate "incidence ratio lab A/lab B" lab 1 -1/ exp alpha=0.05  ;
run;
```

1.12.5 R Code

```
mod1 <- glm(counts~lab,data=salmonella,family="poisson")
summary(mod1)
```

```
Call:
glm(formula = counts ~ lab, family = "poisson", data = salmonella)

Coefficients:
            Estimate Std. Error z value Pr(>|z|)
(Intercept)  4.34019    0.02691  161.29   <2e-16 ***
labB         0.93511    0.03130   29.87   <2e-16 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

(Dispersion parameter for poisson family taken to be 1)

    Null deviance: 1047.539  on 37  degrees of freedom
Residual deviance:   40.345  on 36  degrees of freedom
AIC: 297.69

Number of Fisher Scoring iterations: 4
```

```
library(car)
```

```
Warning: package 'car' was built under R version 4.6.1

Loading required package: carData

Warning: package 'carData' was built under R version 4.6.1


Attaching package: 'car'
```

```
The following object is masked from 'package:dplyr':

    recode

The following object is masked from 'package:purrr':

    some
```

```
Anova(mod1)
```

```
Analysis of Deviance Table (Type II tests)

Response: counts
    LR Chisq Df Pr(>Chisq)
lab   1007.2  1  < 2.2e-16 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
library(performance)
```

```
Warning: package 'performance' was built under R version 4.6.1
```

```
check_overdispersion(mod1)
```

```
Registered S3 method overwritten by 'lme4':
  method           from
  na.action.merMod car

# Overdispersion test

       dispersion ratio =  1.111
  Pearson's Chi-Squared = 40.008
                p-value =  0.297

No overdispersion detected.
```

```
library(emmeans)
```

```
Warning: package 'emmeans' was built under R version 4.6.1
```

```
Welcome to emmeans.
Caution: You lose important information if you filter this package's results.
See '? untidy'
```

```
emm_mod1 <- emmeans(mod1,~lab,type="response", infer=T)
emm_mod1
```

```
 lab  rate   SE  df asymp.LCL asymp.UCL null z.ratio p.value
 A    76.7 2.06 Inf      72.8      80.9    1 161.289 <0.0001
 B   195.4 3.13 Inf     189.4     201.7    1 329.823 <0.0001

Confidence level used: 0.95
Intervals are back-transformed from the log scale
Tests are performed on the log scale
```

```
contrast(emm_mod1,method="pairwise",infer=T)
```

```
 contrast ratio     SE  df asymp.LCL asymp.UCL null z.ratio p.value
 A / B    0.393 0.0123 Inf     0.369     0.417    1 -29.872 <0.0001

Confidence level used: 0.95
Intervals are back-transformed from the log scale
Tests are performed on the log scale
```

1.12.6 Answer

The scaled Pearson chi-square Value/DF = 1.11, which is close to 1. This supports the use of the Poisson distribution to model the salmonella counts.

There is strong evidence of a difference in counts between the two labs ($\chi^2 = 1007$, $p < 0.0001$). The incidence rate ratio is 0.39 (95% CI: 0.37, 0.42), which means there is 61% lower average counts in Lab A compared to Lab B.

1.13 Exercise

The file urgentcare_queue.csv includes simulated queueing data from urgent care clinics located at two cities (A and B). The number of people in queue at the time when the urgent care clinics open is recorded in the variable "queue". The "Medicaid" variable indicates whether the clinic accepts Medicaid or not.

- Use an interaction model to test whether the location and Medicaid acceptance affect the length of the queue of urgent care clinics at the start of the workday. Is the Poisson distribution a good fit for the data? Is the interaction significant at the 0.10 level?
- If the interaction is non-significant, remove the interaction term and re-run the analysis to estimate the incidence rate ratios between the queues in urgent care clinics in City A and B. Use 0.05 as the significance level.

1.14 Exercise

The file urgentcare_queue.csv includes simulated queueing data from urgent care clinics located at two cities (A and B). The number of people in queue at the time when the urgent care clinics open is recorded in the variable "queue". The "Medicaid" variable indicates whether the clinic accepts Medicaid or not.

- Use an interaction model to test whether the location and Medicaid acceptance affect the length of the queue of urgent care clinics at the start of the workday. Is the Poisson distribution a good fit for the data? Is the interaction significant at the 0.10 level?
- If the interaction is non-significant, remove the interaction term and re-run the analysis to estimate the incidence rate ratios between the queues in urgent care clinics in City A and B. Use 0.05 as the significance level.

1.14.1 SAS Code

```
proc genmod data=work.urgent ;
class city medicaid;
model queue = city|medicaid / dist=poisson link=log type3  ;
lsmeans city*medicaid/ilink cl ;
effectplot interaction (sliceby=city);
run;

proc genmod data=work.urgent;
class city medicaid;
model queue = city medicaid / dist=poisson link=log type3  ;
lsmeans city / diff=all ilink cl;
lsmeans medicaid/ diff=all exp cl;
estimate "IRR between City A and B" city 1 -1;
run;
```

1.14.2 R Code

```
urgent <- read.csv("urgentcare_queue.csv",stringsAsFactors = TRUE)
glimpse(urgent)
```

```
Rows: 80
Columns: 3
$ city     <fct> A, A, A, A, A, A, A, A, A, A, A, A, A, A, A, A, A, A, A, A, A~
$ medicaid <fct> Yes, Yes, Yes, No, Yes, No, No, Yes, Yes, Yes, Yes, Yes, Yes,~
$ queue    <int> 4, 6, 4, 4, 6, 3, 6, 2, 3, 3, 3, 2, 6, 8, 7, 7, 5, 4, 7, 5, 6~
```

```
library(performance)
library(car)
library(emmeans)

mod_interaction <- glm(queue~city * medicaid, data=urgent, family="poisson")

check_overdispersion(mod_interaction)
```

```
# Overdispersion test

       dispersion ratio =  0.911
  Pearson's Chi-Squared = 69.200
              p-value =  0.697

No overdispersion detected.
```

```
Anova(mod_interaction)
```

```
Analysis of Deviance Table (Type II tests)

Response: queue
             LR Chisq Df Pr(>Chisq)
city          133.076  1     <2e-16 ***
medicaid        1.332  1     0.2485
city:medicaid   0.613  1     0.4336
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```r
mod1 <- glm(queue~city + medicaid, data=urgent, family="poisson")
summary(mod1)
```

```
Call:
glm(formula = queue ~ city + medicaid, family = "poisson", data = urgent)

Coefficients:
            Estimate Std. Error z value Pr(>|z|)
(Intercept)  1.68311    0.10370  16.230   <2e-16 ***
cityB        0.92258    0.08423  10.953   <2e-16 ***
medicaidYes -0.10868    0.09318  -1.166    0.243
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

(Dispersion parameter for poisson family taken to be 1)

    Null deviance: 205.466  on 79  degrees of freedom
Residual deviance:  70.076  on 77  degrees of freedom
AIC: 384.42

Number of Fisher Scoring iterations: 4
```

```r
check_overdispersion(mod1)
```

```
# Overdispersion test

       dispersion ratio =  0.911
  Pearson's Chi-Squared = 70.132
              p-value =  0.697

No overdispersion detected.
```

```r
Anova(mod1)
```

```
Analysis of Deviance Table (Type II tests)

Response: queue
         LR Chisq Df Pr(>Chisq)
city      133.076  1     <2e-16 ***
```

```
medicaid    1.332  1     0.2485
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
emm_city <- emmeans(mod1,~city,type="response",infer=T)

emm_city
```

```
 city rate    SE  df asymp.LCL asymp.UCL null z.ratio p.value
 A     5.1 0.392 Inf      4.38      5.93    1  21.198 <0.0001
 B    12.8 0.667 Inf     11.58     14.20    1  49.076 <0.0001

Results are averaged over the levels of: medicaid
Confidence level used: 0.95
Intervals are back-transformed from the log scale
Tests are performed on the log scale
```

```
contrast(emm_city, method="pairwise",infer=T)
```

```
 contrast ratio     SE  df asymp.LCL asymp.UCL null z.ratio p.value
 A / B    0.397 0.0335 Inf     0.337     0.469    1 -10.953 <0.0001

Results are averaged over the levels of: medicaid
Confidence level used: 0.95
Intervals are back-transformed from the log scale
Tests are performed on the log scale
```

```
emm_med <- emmeans(mod1,~medicaid,type="response",infer=T)
emm_med
```

```
 medicaid rate    SE  df asymp.LCL asymp.UCL null z.ratio p.value
 No       8.54 0.728 Inf      7.22     10.09    1  25.159 <0.0001
 Yes      7.66 0.354 Inf      6.99      8.38    1  44.053 <0.0001

Results are averaged over the levels of: city
Confidence level used: 0.95
Intervals are back-transformed from the log scale
Tests are performed on the log scale
```

```r
contrast(emm_city, method="pairwise",infer=T)
```

```
 contrast ratio     SE  df asymp.LCL asymp.UCL null z.ratio p.value
 A / B    0.397 0.0335 Inf     0.337     0.469    1 -10.953 <0.0001

Results are averaged over the levels of: medicaid
Confidence level used: 0.95
Intervals are back-transformed from the log scale
Tests are performed on the log scale
```

1.14.3 Answer

The Value/DF of the Scaled Pearson χ^2 (Interaction Model: 0.911) is close to 1, which means the Poisson distribution assumption is appropriate. There is no evidence of an interaction between city and medicaid ($\chi^2 = 0.61$, p=0.43) at the 0.10 level.

The Value/DF of the Scaled Pearson χ^2 for the main effects model is also close to 1 (0.911). The estimated incidence rate of queues in City A is 60% less than that in City B (IRR = 0.40 (95% CI: 0.34, 0.47))

1.15 Example: Offset

The table below includes counts of melanoma cases reported in 1969-1971 for white males in two areas. The totals are the sizes of the estimated populations at risk. The researchers were interested in whether the incidence densities varied across age groups and region.

- Use a main effect model to test for the effect of region and age on the mean number of Melanoma cases among White males reported in 1969-1971.
- Use contrasts to compare the incidence rates for <35 and 75+ age groups.

1.15.1 Table

Table 12.2 New Melanoma Cases Among White Males: 1969–1971

Region	Age Group	Cases	Total
Northern	< 35	61	2880262
Northern	35–44	76	564535
Northern	45–54	98	592983
Northern	55–64	104	450740
Northern	65–74	63	270908
Northern	> 75	80	161850
Southern	< 35	64	1074246
Southern	35–44	75	220407
Southern	45–54	68	198119
Southern	55–64	63	134084
Southern	65–74	45	70708
Southern	> 75	27	34233

1.15.2 SAS Data Set

```
data melanoma;
input age $ region $ cases total;
ltotal=log(total); *calculates the logarithm of the total;                    ①
datalines;
35-44 south 75 220407
45-54 south 68 198119
55-64 south 63 134084
65-74 south 45 70708
75+ south 27 34233
<35 south 64 1074246
35-44 north 76 564535
45-54 north 98 592983
55-64 north 104 450740
65-74 north 63 270908
75+ north 80 161850
<35 north 61 2880262
;
```

① The `ltotal` variable performs the logarithmic transformation of the offset variable, which is the total number of tests.

1.15.3 R Data Set

```
melanoma <- expand.grid(Age_Group = c("<35","35-44","45-54","55-64","65-74",">75"),
                        Region = c("Northern","Southern"))

melanoma$Cases <- c(61,76,98,104,63,80,64,75,68,63,45,27)
melanoma$Total <- c(2880262,564535,592983,450740,270908,161850,1074246,220407,198119,134084,
melanoma
```

```
   Age_Group   Region Cases   Total
1        <35 Northern    61 2880262
2      35-44 Northern    76  564535
3      45-54 Northern    98  592983
4      55-64 Northern   104  450740
5      65-74 Northern    63  270908
6        >75 Northern    80  161850
7        <35 Southern    64 1074246
8      35-44 Southern    75  220407
9      45-54 Southern    68  198119
10     55-64 Southern    63  134084
11     65-74 Southern    45   70708
12       >75 Southern    27   34233
```

1.15.4 SAS Code

```
proc genmod data=melanoma;
class region (ref='north') age (ref='<35');
model cases = age region
/ dist=poisson link=log offset=ltotal type3 cl ; *the offset should be in the LOGARITHMIC SC
estimate "75+ vs <35" age 0 0 0 0 1 -1/exp;
lsmeans age/ ilink cl diff = all  adjust=tukey;
lsmeans region /ilink cl diff = all ;
run;
```

1.15.5 R Code

```
mod1 <- glm(Cases~Age_Group + Region, data=melanoma,family="poisson",offset=log(Total))
summary(mod1)
```

```
Call:
glm(formula = Cases ~ Age_Group + Region, family = "poisson",
    data = melanoma, offset = log(Total))

Coefficients:
                Estimate Std. Error z value Pr(>|z|)
(Intercept)    -10.65831    0.09518 -111.97   <2e-16 ***
Age_Group35-44   1.79737    0.12093   14.86   <2e-16 ***
Age_Group45-54   1.91309    0.11844   16.15   <2e-16 ***
Age_Group55-64   2.24180    0.11834   18.94   <2e-16 ***
Age_Group65-74   2.36572    0.13152   17.99   <2e-16 ***
Age_Group>75     2.94468    0.13205   22.30   <2e-16 ***
RegionSouthern   0.81948    0.07103   11.54   <2e-16 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

(Dispersion parameter for poisson family taken to be 1)

    Null deviance: 895.8197  on 11  degrees of freedom
Residual deviance:   6.2149  on  5  degrees of freedom
AIC: 92.44

Number of Fisher Scoring iterations: 4
```

```
check_overdispersion(mod1)
```

```
# Overdispersion test

       dispersion ratio = 1.223
  Pearson's Chi-Squared = 6.115
                p-value = 0.295

No overdispersion detected.
```

```r
Anova(mod1)
```

```
Analysis of Deviance Table (Type II tests)

Response: Cases
          LR Chisq Df Pr(>Chisq)
Age_Group   796.74  5  < 2.2e-16 ***
Region      124.22  1  < 2.2e-16 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```r
emm_age <- emmeans(mod1,~Age_Group, infer=T, type="response")
emm_age
```

```
 Age_Group  rate     SE  df asymp.LCL asymp.UCL null z.ratio p.value
 <35        10.1  0.904 Inf      8.48      12.0    1  25.841 <0.0001
 35-44      60.9  4.960 Inf     51.95      71.5    1  50.486 <0.0001
 45-54      68.4  5.320 Inf     58.74      79.7    1  54.336 <0.0001
 55-64      95.0  7.380 Inf     81.62     110.7    1  58.623 <0.0001
 65-74     107.6 10.400 Inf     89.02     130.0    1  48.402 <0.0001
 >75       191.9 18.700 Inf    158.56     232.4    1  53.934 <0.0001

Results are averaged over the levels of: Region
Confidence level used: 0.95
Intervals are back-transformed from the log scale
Tests are performed on the log scale
```

```r
contrast_list <- list(contrast1 = c(-1,0,0,0,0,1))

contrast(emm_age,method=contrast_list, infer=T)
```

```
 contrast  ratio   SE  df asymp.LCL asymp.UCL null z.ratio p.value
 contrast1    19 2.51 Inf      14.7      24.6    1  22.300 <0.0001

Results are averaged over the levels of: Region
Confidence level used: 0.95
Intervals are back-transformed from the log scale
Tests are performed on the log scale
```

1.15.6 Answer

Note that the Scaled Pearson Chi-Square is 1.24, which is slightly further away from 1 and should be examined more closely using methods we will discuss later in the chapter. The overdispersion test using the `performance` package yielded a p-value of 0.295, which indicates there is no evidence of overdispersion. This result suggests we can continue analyzing the data using the Poisson distribution.

There is strong evidence of an age effect ($\chi^2 = 797$, p <0.0001) and region ($\chi^2 = 124.22$, p < 0.001).

The incidence rate of Melanoma in the 75+ age group is 19 times (95% CI: 14.67, 24.62) that in the <35 age group.

1.16 Exercise

The file resident_hours.csv contains the annual count of patients assisted by medical residents at a hospital in Asia. Researchers wanted to test whether the patient assistance rate (#patients per hour) has decreased throughout the years. The number of hours spent on duty is listed as the "time" variable— with "ltime" being the logarithm of the time variable.

- Is there an overall effect of year?
- Calculate the change in assistance rate from 2021 to 2022, and from 2022 to 2023.

> **Note**
>
> Because we are performing multiple contrasts simultaneously, we need to account for Type I error inflation. Use the BOnferroni correction.

1.17 Exercise

The file resident_hours.csv contains the annual count of patients assisted by medical residents at a hospital in Asia. Researchers wanted to test whether the patient assistance rate (#patients per hour) has decreased throughout the years. The number of hours spent on duty is listed as the "time" variable— with "ltime" being the logarithm of the time variable.

1.17.1 SAS Code

```
proc genmod data=resident;
class year;
model counts = year
/ dist=poisson link=log offset=ltime type3 cl; *the offset should be in the LOGARITHMIC SCALE
lsmeans year/ ilink cl diff=all adjust=bonferroni;
estimate "2021 vs 2022" year 1 -1 0;
estimate "2022 vs 2023" year 0 1 -1 ;
run;
```

1.17.2 R Code

```
resident <- read.csv("resident_hours.csv")
glimpse(resident)
```

```
Rows: 38
Columns: 4
$ year   <int> 2021, 2021, 2021, 2021, 2021, 2021, 2021, 2021, 2021, 2021, 202~
$ counts <int> 2864, 2907, 2901, 2890, 2966, 2896, 2934, 2924, 2912, 2889, 203~
$ time   <int> 3086, 3145, 3163, 3049, 3136, 3093, 3137, 3089, 3089, 3109, 313~
$ ltime  <dbl> 8.034631, 8.053569, 8.059276, 8.022569, 8.050703, 8.036897, 8.0~
```

```
resident <- resident |>
  mutate(year = as.factor(year))
```

```
mod1 <- glm(counts~year, data=resident, family="poisson",offset=log(time))
summary(mod1)
```

```
Call:
glm(formula = counts ~ year, family = "poisson", data = resident,
    offset = log(time))

Coefficients:
             Estimate Std. Error z value Pr(>|z|)
(Intercept) -0.066925   0.005864  -11.41   <2e-16 ***
year2022    -0.373730   0.008527  -43.83   <2e-16 ***
year2023    -1.255312   0.010705 -117.26   <2e-16 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
(Dispersion parameter for poisson family taken to be 1)

    Null deviance: 15918.861  on 37  degrees of freedom
Residual deviance:    27.912  on 35  degrees of freedom
AIC: 383.2

Number of Fisher Scoring iterations: 3
```

```
check_overdispersion(mod1)
```

```
# Overdispersion test

       dispersion ratio =  0.795
  Pearson's Chi-Squared = 27.838
              p-value =    0.8

No overdispersion detected.
```

```
Anova(mod1)
```

```
Analysis of Deviance Table (Type II tests)

Response: counts
     LR Chisq Df Pr(>Chisq)
year    15891  2  < 2.2e-16 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
emm_year <- emmeans(mod1,~year,type="response",infer=T)
emm_year
```

```
 year rate    SE  df asymp.LCL asymp.UCL null  z.ratio p.value
 2021 2914 17.10 Inf      2881      2948    1 1360.439 <0.0001
 2022 2005 12.40 Inf      1981      2030    1 1228.311 <0.0001
 2023  831  7.44 Inf       816       845    1  750.526 <0.0001

Confidence level used: 0.95
Intervals are back-transformed from the log scale
Tests are performed on the log scale
```

```r
contrast_list <- list(contrast_2021v2022 = c(1,-1,0),
                      contrast_2022v2023 = c(0,1,-1))

contrast(emm_year,method=contrast_list,infer=T,adjust="bonferroni")
```

```
 contrast           ratio     SE  df asymp.LCL asymp.UCL null z.ratio p.value
 contrast_2021v2022  1.45 0.0124 Inf      1.43      1.48    1  43.831 <0.0001
 contrast_2022v2023  2.41 0.0263 Inf      2.36      2.47    1  80.972 <0.0001

Confidence level used: 0.95
Conf-level adjustment: bonferroni method for 2 estimates
Intervals are back-transformed from the log scale
P value adjustment: bonferroni method for 2 tests
Tests are performed on the log scale
```

1.17.3 Answer

The Scaled Pearson χ^2 is 0.8, which is slightly lower than 1. In this course, we are only concerned if the Scaled Pearson X2 is above 1, so we can say that the Poisson distribution is appropriate to use.

There is an overall effect of year ($ ^2 = 15891, p < 0.0001).

When looking at multiple pairwise differences, it is important to adjust for multiplicity, hence we added `adjust=bonferroni`. After applying the Tukey-Kramer adjustment, we estimate that the patient assistance rate is 45% times higher than that in 2021 compared to 2022 (95% CI: 43%, 48%). Additionally, the patient assistance rate is 141% (95% CI: 136%, 147%) higher in 2022 compared to 2023.

1.18 Overdispersion

- Recall: We use the Scaled chi-square /df value to determine whether the Poisson distribution is a good fit to the data.
  - What if the Scaled chi-square/df » 1? Overdispersion
- Overdispersion occurs when the observed variance is larger than the variance accounted for in the model.
  - Recall: Poisson distribution mean = Poisson distribution variance = μ.
- Underdispersion also occurs: Scaled chi-square/df «1
  - Will not be discussed in this course.

1.19 Overdispersion: What to do?

1.19.1 Quasi-Poisson Approximation

If we want to continue using the Poisson distribution, we can introduce a scaling factor ϕ such that the variance is assumed to be $\phi\mu$. ϕ is then approximated using the data.

1.19.2 Negative Binomial Model

We can use alternative distributions like the negative binomial distribution. The negative binomial can be reparameterized by multiplying a scaling factor to account for additional dispersion in the data. Specifically, the variance is assumed to be $\phi(1 + k\mu)$.

1.19.3 Mixed models

Adding random effects to the model can account for the excess variability that is unaccounted for by the current model.

1.19.4 Zero-Inflation

A potential cause for overdispersion is the presence of excess zeroes. These zeroes can exist due to the nature of the process, method of measurement, or to external factors. Zero-inflated Poisson, Zero-inflated Negative Binomial, and hurdle models are used to account for these zeroes. The approach taken depends on the cause of excess zeroes.

1.20 Negative Binomial Model

The negative binomial model arises as a continuous mixture of Poisson distributions where the rate parameters follow a Gamma distribution. Hence, using the negative binomial model is also referred to as the Poisson-Gamma reparameterization.

> **Tip**
>
> Biological processes tend to agree with negative binomial models more than Poisson regression. Rule of thumb is to use Poisson unless the data is overdispersed, in which you use the negative binomial model.

1.21 Overdispersion: SAS

- PROC GENMOD can handle both Poisson and Negative Binomial analyses
  - DIST=NB specifies a negative binomial distribution.
  - Dispersion now replaces Scale in the output
  - We can also adjust the scale using the option `scale=pearson` in the `MODEL` statement (Quasi-Poisson approach).

1.22 Overdispersion: `R`

Once `check_overdispersion()` yields a significant result for overdispersion, we can modify the distribution used by `glm()` to analyze the data.

> **Note**
>
> `family="quasipoisson"` uses the quasi-Poisson distribution.

> **Important**
>
> The negative binomial GLM can be implemented using the `glm.nb()` function in the `MASS` package.

1.23 Example

The data set OVERDISPERSION shows count data from different locations and treatments. Suppose we want to test for a treatment effect.

1.23.1 SAS Data Set

```
data overdispersion;
 input location trt count;
datalines;
1 1 2
1 2 3
1 3 0
1 4 2
2 1 26
2 2 14
2 5 51
```

```
2 6 16
3 1 76
3 3 44
3 5 77
3 7 3
4 1 0
4 4 3
4 6 5
4 7 12
5 2 12
5 3 13
5 6 22
5 7 1
6 2 49
6 4 54
6 5 205
6 7 23
7 3 3
7 4 4
7 5 15
7 6 3
;
```

1.23.2 R Data Set

```
over <- read.csv("overdispersion.csv")
glimpse(over)
```

```
Rows: 28
Columns: 3
$ location <int> 1, 1, 1, 1, 2, 2, 2, 2, 3, 3, 3, 3, 4, 4, 4, 4, 5, 5, 5, 5, 6~
$ trt      <int> 1, 2, 3, 4, 1, 2, 5, 6, 1, 3, 5, 7, 1, 4, 6, 7, 2, 3, 6, 7, 2~
$ count    <int> 2, 3, 0, 2, 26, 14, 51, 16, 76, 44, 77, 3, 0, 3, 5, 12, 12, 1~
```

```
over <- over |>
  mutate(location = as.factor(location),
         trt = as.factor(trt))
```

1.23.3 SAS Codes

```
/*Example of overdispersion*/
proc genmod data=overdispersion;
class location trt;
model count=trt location/type3 dist=poisson;
lsmeans trt/ilink cl;
lsmeans location/ilink cl;
estimate "trt 2 vs trt 4" trt 0 1 0 0 0 -1 0;
run;

/*Pearson adjustment*/
proc genmod data=overdispersion;
class location trt;
model count=trt location/type3 dist=poisson scale=pearson;
lsmeans trt/ilink cl;
lsmeans location/ilink cl;
estimate "trt 2 vs trt 6" trt 0 1 0 0 0 -1 0;
run;


/*Negative Binomial*/
proc genmod data=overdispersion;
class location trt;
model count=trt location/type3 dist=nb ;
lsmeans trt/ilink cl;
lsmeans location/ilink cl;
estimate "trt 2 vs trt 6" trt 0 1 0 0 0 -1 0;
run;
```

1.23.4 R Code

We start with the Poisson model first and check for overdispersion

```
mod_poisson <- glm(count~trt+location,
            data=over,
            family="poisson")

library(car)
Anova(mod_poisson)
```

```
Analysis of Deviance Table (Type II tests)

Response: count
         LR Chisq Df Pr(>Chisq)
trt        328.33  6  < 2.2e-16 ***
location   598.14  6  < 2.2e-16 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
library(emmeans)
emm_pois <- emmeans(mod_poisson,~trt,type="response",infer=T)
emm_pois
```

```
 trt  rate    SE  df asymp.LCL asymp.UCL null z.ratio p.value
 1   20.34 2.460 Inf     16.04     25.78    1  24.897 <0.0001
 2    8.14 1.120 Inf      6.21     10.67    1  15.202 <0.0001
 3   12.31 1.810 Inf      9.22     16.43    1  17.023 <0.0001
 4    8.28 1.250 Inf      6.16     11.15    1  13.967 <0.0001
 5   30.17 2.910 Inf     24.96     36.46    1  35.261 <0.0001
 6   12.55 2.060 Inf      9.10     17.32    1  15.404 <0.0001
 7    3.42 0.607 Inf      2.42      4.84    1   6.931 <0.0001

Results are averaged over the levels of: location
Confidence level used: 0.95
Intervals are back-transformed from the log scale
Tests are performed on the log scale
```

```
library(performance)

check_overdispersion(mod_poisson)
```

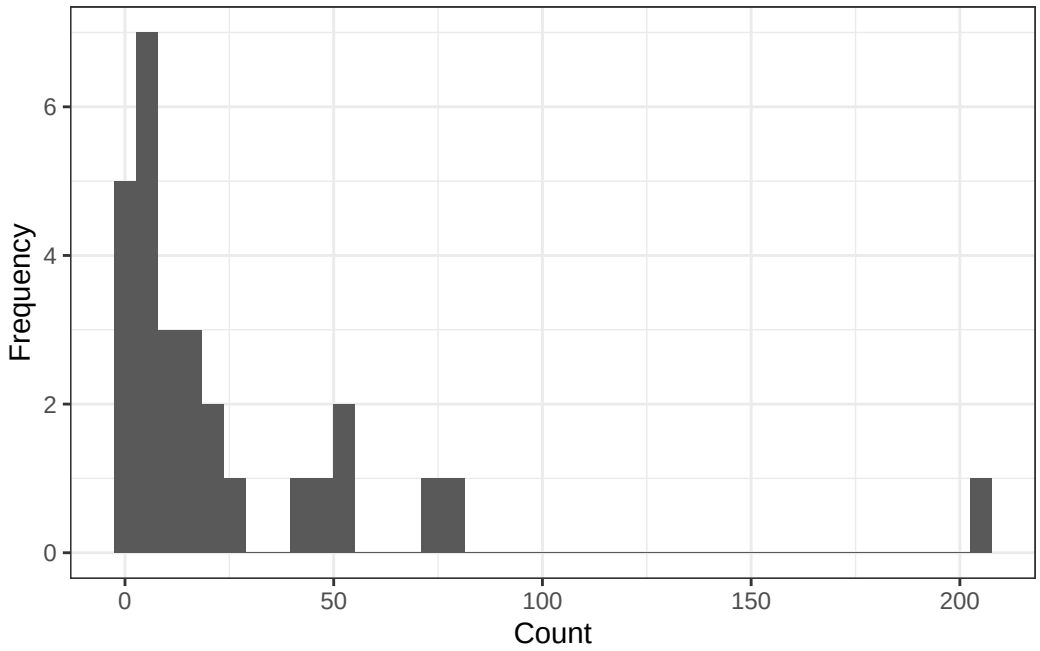
```
# Overdispersion test

       dispersion ratio =   7.255
  Pearson's Chi-Squared = 108.820
              p-value = < 0.001

Overdispersion detected.
```

Overdispersion is detected. First, we check whether the overdispersion is due to excess zeroes by creating a histogram of counts.

```r
over |>
  ggplot(aes(x=count)) +
  geom_histogram(bins=40) +
  theme_bw() +
  labs(x="Count", y="Frequency")
```



While there are a lot of zeroes, there are also nonzero values that can be driving the overdispersion.

```r
mod1 <- glm(count~trt+location,
            data=over,
            family="quasipoisson")
summary(mod1)
```

```
Call:
glm(formula = count ~ trt + location, family = "quasipoisson",
    data = over)

Coefficients:
            Estimate Std. Error t value Pr(>|t|)
(Intercept)   1.0651     1.0402   1.024  0.32210
trt2         -0.9156     0.4528  -2.022  0.06137 .
```

```
trt3         -0.5023     0.4645  -1.081  0.29660
trt4         -0.8981     0.4883  -1.839  0.08577 .
trt5          0.3943     0.3398   1.160  0.26412
trt6         -0.4824     0.5245  -0.920  0.37223
trt7         -1.7829     0.5331  -3.345  0.00443 **
location2     2.3547     1.0615   2.218  0.04238 *
location3     3.0525     1.0417   2.930  0.01034 *
location4     1.1454     1.1939   0.959  0.35258
location5     2.2234     1.1099   2.003  0.06356 .
location6     3.8372     1.0471   3.665  0.00230 **
location7     1.0181     1.1641   0.875  0.39560
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

(Dispersion parameter for quasipoisson family taken to be 7.254675)

    Null deviance: 1171.998  on 27  degrees of freedom
Residual deviance:   78.956  on 15  degrees of freedom
AIC: NA

Number of Fisher Scoring iterations: 6
```

```
Anova(mod1)
```

```
Analysis of Deviance Table (Type II tests)

Response: count
         LR Chisq Df Pr(>Chisq)
trt        45.258  6  4.158e-08 ***
location   82.449  6  1.114e-15 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
emmeans(mod1,~trt,infer=T,type="response")
```

```
 trt  rate   SE  df asymp.LCL asymp.UCL null z.ratio p.value
 1   20.34 6.63 Inf     10.74     38.52    1   9.244 <0.0001
 2    8.14 3.02 Inf      3.93     16.86    1   5.644 <0.0001
 3   12.31 4.89 Inf      5.65     26.81    1   6.320 <0.0001
 4    8.28 3.38 Inf      3.73     18.42    1   5.186 <0.0001
 5   30.17 7.85 Inf     18.11     50.24    1  13.091 <0.0001
```

```
 6   12.55 5.55 Inf      5.27     29.88    1   5.719 <0.0001
 7    3.42 1.63 Inf      1.34      8.72    1   2.573  0.0101

Results are averaged over the levels of: location
Confidence level used: 0.95
Intervals are back-transformed from the log scale
Tests are performed on the log scale
```

> **Note**
>
> The dispersion parameter of the quasiPoisson model is the same as the dispersion ratio in the overdispersion test.

For the negative binomial model,

```
library(MASS)
```

```
Attaching package: 'MASS'

The following object is masked from 'package:dplyr':

    select
```

```
mod_nb <- glm.nb(count~trt+location,
            data=over,
            control = glm.control(maxit = 1000)) # for convergence issues


summary(mod_nb)
```

```
Call:
glm.nb(formula = count ~ trt + location, data = over, control = glm.control(maxit = 1000),
    init.theta = 5.695838947, link = log)

Coefficients:
            Estimate Std. Error z value Pr(>|z|)
(Intercept)   0.8616     0.4893   1.761 0.078238 .
trt2         -0.3832     0.4041  -0.948 0.343030
```

```
trt3         -0.3606     0.4148  -0.869 0.384771
trt4         -0.5063     0.4320  -1.172 0.241132
trt5          0.8106     0.3674   2.206 0.027378 *
trt6         -0.2381     0.3976  -0.599 0.549200
trt7         -1.1752     0.4022  -2.922 0.003479 **
location2     2.2499     0.5107   4.406 1.06e-05 ***
location3     2.9928     0.5052   5.924 3.15e-09 ***
location4     1.3887     0.5393   2.575 0.010029 *
location5     2.0278     0.5256   3.858 0.000114 ***
location6     3.5463     0.5075   6.987 2.80e-12 ***
location7     0.8436     0.5501   1.533 0.125180
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

(Dispersion parameter for Negative Binomial(5.6958) family taken to be 1)

    Null deviance: 223.862  on 27  degrees of freedom
Residual deviance:  37.766  on 15  degrees of freedom
AIC: 213.74

Number of Fisher Scoring iterations: 1

              Theta:  5.70
          Std. Err.:  3.09

 2 x log-likelihood:  -185.735
```

```r
# Recreates SAS output
mod_nb_loc <- glm.nb(count~location,
            data=over,
            control = glm.control(maxit = 1000))
anova(mod_nb_loc,mod_nb)
```

```
Likelihood ratio tests of Negative Binomial Models

Response: count
           Model    theta Resid. df    2 x log-lik.   Test    df LR stat.
1       location 1.854654        21       -201.0688
2 trt + location 5.695839        15       -185.7351 1 vs 2     6 15.33371
     Pr(Chi)
1
2 0.01781411
```

1.23.5 Answer

In the Naive Poisson approach, the Scaled Pearson X2 Value/DF was 7.25, which is significantly higher than 1. We can choose to add `scale=Pearson` to include a scale factor, making the distribution a quasi-Poisson distribution. The Poisson and quasi-Poisson have the same Least Squares Means (LSMeans) estimates, but have different standard errors. Note that the scale introduces a scaling factor to the variance estimate in the form of the dispersion parameter, which affects the standard errors. For both models, there is a significant treatment effect (χ^2_6=45.26, p $<$ 0.0001)

The negative binomial model also includes the dispersion parameter in the results, and has different LSMeans estimates as the Poisson. (LR $\chi^2_6 = 15.33$, p=0.0178).

1.24 Exercise

The file adverse.csv includes the counts of adverse events recorded for subjects in each of the four experimental treatment regimes for ovarian cancer. The subjects were recruited from three clinics. The researchers want to know if the four treatments differ in incidence density rate of adverse events. (Hint: Is there overdispersion? Did the resulting model converge?)

1.25 Exercise

The file adverse.csv includes the counts of adverse events recorded for subjects in each of the four experimental treatment regimes for ovarian cancer. The subjects were recruited from three clinics. The researchers want to know if the four treatments differ in incidence density rate of adverse events. (Hint: Is there overdispersion? Did the resulting model converge?)

1.25.1 SAS Code

```
proc genmod data=adverse;
class clinic treatment;
model count = clinic treatment/ type3 dist=poisson;
lsmeans treatment/ilink cl;
run;

proc genmod data=adverse;
class clinic treatment;
model count = clinic treatment/ type3 dist=poisson scale=pearson;
lsmeans treatment/ilink cl;
run;
```

```
proc genmod data=adverse;
class clinic treatment;
model count = clinic treatment/ type3 dist=nb;
lsmeans treatment/ilink cl;
run;
```

1.25.2 R Code

```
adverse <- read.csv("adverse.csv")
glimpse(adverse)
```

```
Rows: 12
Columns: 3
$ Clinic    <int> 1, 2, 3, 1, 2, 3, 1, 2, 3, 1, 2, 3
$ Treatment <chr> "a", "a", "a", "b", "b", "b", "c", "c", "c", "d", "d", "d"
$ Count     <int> 9, 4, 12, 24, 11, 17, 2, 14, 8, 15, 5, 23
```

```
adverse <- adverse |>
  mutate(Clinic = as.factor(Clinic))
```

```
# Poisson model

mod_poisson <- glm(Count~Treatment + Clinic, data=adverse,family="poisson")
check_overdispersion(mod_poisson)
```

```
# Overdispersion test

       dispersion ratio =   4.248
  Pearson's Chi-Squared =  25.487
              p-value = < 0.001

Overdispersion detected.
```

```
Anova(mod_poisson)
```

```
Analysis of Deviance Table (Type II tests)
```

```
Response: Count
          LR Chisq Df Pr(>Chisq)
Treatment  15.8295  3   0.001229 **
Clinic      7.4103  2   0.024597 *
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```r
# quasiPoisson model

mod_q<- glm(Count~Treatment + Clinic, data=adverse,family="quasipoisson")
Anova(mod_q)
```

```
Analysis of Deviance Table (Type II tests)

Response: Count
          LR Chisq Df Pr(>Chisq)
Treatment   3.7265  3     0.2926
Clinic      1.7445  2     0.4180
```

```r
# Negative Binomial
mod_nb <- glm.nb(Count~Treatment + Clinic,
                 data=adverse,
                 control = glm.control(maxit = 1000))

summary(mod_nb)
```

```
Call:
glm.nb(formula = Count ~ Treatment + Clinic, data = adverse,
    control = glm.control(maxit = 1000), init.theta = 11.34489479,
    link = log)

Coefficients:
            Estimate Std. Error z value Pr(>|z|)
(Intercept)  2.09946    0.31649   6.634 3.28e-11 ***
Treatmentb   0.75575    0.34626   2.183   0.0291 *
Treatmentc   0.02515    0.37506   0.067   0.9465
Treatmentd   0.52200    0.35354   1.477   0.1398
Clinic2     -0.29491    0.30935  -0.953   0.3404
Clinic3      0.23714    0.28942   0.819   0.4126
---
```

```
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

(Dispersion parameter for Negative Binomial(11.3449) family taken to be 1)

    Null deviance: 24.663  on 11  degrees of freedom
Residual deviance: 13.902  on  6  degrees of freedom
AIC: 85.807

Number of Fisher Scoring iterations: 1

              Theta:  11.3
          Std. Err.:  10.8

 2 x log-likelihood:  -71.807
```

```
check_overdispersion(mod_nb)
```

```
# Overdispersion test (using simulated residuals)

 dispersion ratio = 0.886
          p-value = 0.944

No overdispersion detected.
```

```
# recreating SAS tests

mod_nb_trt <- glm.nb(Count~Treatment ,
                  data=adverse,
                  control = glm.control(maxit = 1000))

mod_nb_cli <- glm.nb(Count~Clinic,
                  data=adverse,
                  control = glm.control(maxit = 1000))

anova(mod_nb,mod_nb_trt)
```

```
Likelihood ratio tests of Negative Binomial Models

Response: Count
               Model     theta Resid. df    2 x log-lik.   Test    df LR stat.
1          Treatment  6.662764         8       -74.31619
```

```
2 Treatment + Clinic 11.344895         6       -71.80741 1 vs 2     2 2.508777
    Pr(Chi)
1
2 0.2852503
```

```
anova(mod_nb,mod_nb_cli)
```

```
Likelihood ratio tests of Negative Binomial Models

Response: Count
               Model     theta Resid. df    2 x log-lik.   Test    df LR stat.
1             Clinic  4.629789         9       -76.94175
2 Treatment + Clinic 11.344895         6       -71.80741 1 vs 2     3 5.134338
    Pr(Chi)
1
2 0.1622204
```

1.25.3 Answer

After implementing the Naive Poisson model, the scaled Pearson X2 Value/DF is 4.25, which is a sign that there is overdispersion. There was no convergence issues in implementing the quasiPoisson and the negative binomial model/

Quasipoisson: There is no significant effect of clinic ($\chi^2_2 = 1.74$, p = 0.42) and treatment ($\chi^2_3 = 3.73$, p = 0.29).

Negative Binomial: There is no significant effect of clinic ($\chi^2_2 = 2.51$, p = 0.29) and treatment ($\chi^2_3 = 5.13$, p = 0.16).

1.26 Exact Poisson Regression: SAS

> **Tip**
>
> For rare events with small number of observed counts, exact methods may be more appropriate. The `GENMOD` procedure can also implement exact Poisson regression analysis using the `exact` statement.

1.27 Example

The table shows the rotavirus vaccine data for both United States and Latin America. We want to investigate if there is a difference between the incidence densities of the vaccine and placebo groups.

1.27.1 Table

Table 12.3 Medical Events for Rotavirus Vaccine Study

	Vaccine		Placebo	
Region	Events	Person Years	Events	Person Years
United States	3	7500	58	7250
Latin America	1	1250	10	1250

1.27.2 SAS Data Set

```
data rotavirus;
input region $ treatment $ counts years_risk @@ ;
log_risk=log(years_risk);
datalines;
US Vaccine 3 7500 US Placebo 58 7250
LA Vaccine 1 1250 LA Placebo 10 1250
;
run;
```

1.27.3 SAS Code

```
/*Not Exact*/
proc genmod data=rotavirus;
class region treatment;
model counts = treatment region / dist=poisson offset= log_risk type3;
lsmeans treatment/ilink cl;

estimate 'Placebo vs Vaccine' treatment 1 -1 /exp;
```

```
/*Exact*/
proc genmod data=rotavirus;
class region treatment /param=ref;
model counts = treatment region / dist=poisson offset= log_risk type3;
lsmeans treatment/ilink cl;
estimate 'Placebo vs Vaccine' treatment 1 /exp;
exact treatment / estimate=odds cltype=exact;
run;
```

1.27.4 Answer

It would be advisable to use the exact incidence rate ratio confidence limits and p-values because of the low count of events in the vaccine group. The incidence rate in the placebo group was 17.5 **Exact 95% CI: (6.53, 66.05)** times that of the vaccine group.

1.28 Extreme Overdispersion: Zero inflation

Oftentimes, you can encounter data sets with an abundance of zeroes especially when the event of interest is rare.

- Behavioral events in students
- Positive cases in vaccine groups
- Insect count in a pesticide-treated plot of land.

> **Warning**
>
> Using a negative binomial distribution or the quasi-Poisson likelihood will not be enough to account for the abundance of zeros.

1.29 Zero-Inflated Poisson/Negative Binomial models

Zero-inflated models account for excess zeroes that are not accounted for by the Poisson or Negative Binomial distributions.

There are two types of zero inflated models:

> **Zero-Inflated Models**
>
> In zero-inflated models, zero or nonzero counts may occur in the "on" part of the process, with the probability of 0, 1, 2, etc., determined by the counting distribution in the "on"

part of the process.
Example: Cars at a traffic intersection, animal sightings.

Hurdle Models

In hurdle models, only nonzero counts may be observed in the "on" part of the process.
Example: Doctor visits, Insurance claims

1.30 Zero-inflated Models

The zero-inflated models must be described in two processes: the on-off process and the counting process.

Process 1

The on-off part is a binary process. The count process is "off" or 0 with probability π and "on" with probability $1-\pi$. We call π the inflation probability.

Process 2

When the system is "on", counts may take on values 0,1,2,... according to a discrete probability distribution such as Poisson or Negative Binomial. This gives rise to Zero-Inflated Poisson (ZIP) models and Zero-Inflated Negative Binomial (ZINB) models.

The resulting probability $P(Y=y)$ where Y is the random variable corresponding to the observations from a zero-inflated model is:

$$P(Y=y) = \begin{cases} \pi + (1-\pi)f(0) & y=0 \\ (1-\pi)f(y) & y=1,2,... \end{cases}$$

where $f(y)$ is the discrete probability distribution corresponding to Poisson/Negative Binomial.

1.31 Hurdle Model

Like the zero-inflated models, hurdle models also have two processes: the on-off process and the counting process.

Process 1

The on-off part is a binary process. The count process is "off" or 0 with probability π and "on" with probability $1-\pi$. We call π the inflation probability. This is the same as the on-off process for zero-inflated models.

Process 2

The "on" process is slightly different. Note that the hurdle model is used when zero counts is NOT possible during the "on" process. Hence, we have to use a zero-truncated model of the distribution function $\frac{f(y)}{1-f(0)}, y = 1, 2, ...$

The resulting probability distribution for the hurdle model is thus:

$$P(Y = y) = \begin{cases} \pi & y = 0 \\ (1-\pi)\frac{f(y)}{1-f(0)} & y = 1, 2, ... \end{cases}$$

1.32 Zero Inflated Modeling in `SAS`

The `GENMOD` procedure has ZIP and ZINB capabilities, though limited.

Note

Specifying `dist=zip` and `dist=zinb` in the `model` statement will implement zero-inflated Poisson and Negative Binomial analyses, respectively.

Tip

The `zeromodel` statement includes variables that can introduce group-specific zero inflation.

Warning

The `GENMOD` procedure cannot implement the Hurdle model.

1.33 Zero-Inflated Modeling in `R`

The `pscl` package includes the `zeroinfl()` function that can implement zero-inflated Poisson and Negative Binomial models. The `hurdle()` function can be used to implement hurdle models.

> **Note**
>
> To perform the likelihood ratio test, we use the `lrtest()` function from the `lmtest()` package. `anova()` does not work with `zeroinfl()`.

1.34 Example

The data set `counts` contain simulated data from an experiment investigating the association between exposure level and toxin counts (also in `counts.csv`. There were n_i samples taken at the ith sites, ranging from 2 to 14 samples, at 18 sites. Determine if there is evidence of zero-inflation. Test for an association between exposure level and toxin counts.

1.34.1 Data Set

```
data counts;
   input ni @@;
   site = _n_;
   do sample_id=1 to ni;
     input exposure_level toxin_count @@; output;
   end;
   datalines;
  6  2 0 82 5 33 0 15 2 35 0 79 1
  9 18 0 64 0 80 2  0 0 58 0  7 0 81 0 22 3 50 0
  2 34 0 95 0
 13 28 1 31 0 63 2 14 0 74 0 44 0 75 3 65 0 74 1 84 5 57 0 29 0 41 0
  9 42 0  8 0 91 0 20 0 23 0 22 0 96 4 83 3 56 0
  3 64 0 64 1 15 0
  4  5 0 73 2 50 2 13 1
  2  0 0 41 0
  5 83 7 98 1 11 1 28 0 18 0
  7 91 0 25 1 51 4 20 0 61 1 34 0 33 2
 14 60 0 87 0 94 0 29 0 41 0 78 0 50 0 37 0 15 0 39 0 22 0 82 0 93 0 3 0
 12 68 0 26 1 19 0 60 1 93 3 65 0 16 0 79 0 14 0  3 1 90 0 28 3
  8 34 1 44 0 62 3 21 0  7 0 17 1  0 2 49 0
 13 11 0 27 2 16 1 12 3 52 1 55 0  2 6 89 5 31 5 28 3 51 5 54 13 64 0
  9  3 0 36 0 57 0 77 0 41 0 39 0 55 0 57 0 88 1
  7  2 0 80 0 41 1 20 0  2 0 27 0 40 0
 13 59 0 96 2 47 1 64 0 18 0 30 0 37 0 36 1 69 0 78 1 47 1 86 0 88 0
 12 84 6 60 1 33 1 92 0 38 4  6 0 43 3 13 2 18 0 51 0 50 4 68 0
```

```
;
```

1.34.2 R Code

```
counts <- read.csv("counts.csv")
glimpse(counts)
```

```
Rows: 148
Columns: 5
$ ni             <int> 6, 6, 6, 6, 6, 6, 9, 9, 9, 9, 9, 9, 9, 9, 9, 2, 2, 13, ~
$ site           <int> 1, 1, 1, 1, 1, 1, 2, 2, 2, 2, 2, 2, 2, 2, 2, 3, 3, 4, 4~
$ sample_id      <int> 1, 2, 3, 4, 5, 6, 1, 2, 3, 4, 5, 6, 7, 8, 9, 1, 2, 1, 2~
$ exposure_level <int> 2, 82, 33, 15, 35, 79, 18, 64, 80, 0, 58, 7, 81, 22, 50~
$ toxin_count    <int> 0, 5, 0, 2, 0, 1, 0, 0, 2, 0, 0, 0, 0, 3, 0, 0, 0, 1, 0~
```

```
mod_poisson <- glm(toxin_count~exposure_level,data=counts,family="poisson")
library(performance)
check_zeroinflation(mod_poisson)
```

```
# Check for zero-inflation

   Observed zeros: 92
  Predicted zeros: 58
            Ratio: 0.63

Model is underfitting zeros (probable zero-inflation).
```

There is evidence of unaccounted zeroes, hence we need to use zero-inflated models.

```
library(pscl)
```

```
Warning: package 'pscl' was built under R version 4.6.1

Classes and Methods for R originally developed in the
Political Science Computational Laboratory
Department of Political Science
Stanford University (2002-2015),
by and under the direction of Simon Jackman.
hurdle and zeroinfl functions by Achim Zeileis.
```

```r
library(lmtest)
```

```
Warning: package 'lmtest' was built under R version 4.6.1
```

```
Loading required package: zoo
```

```
Warning: package 'zoo' was built under R version 4.6.1
```

```

Attaching package: 'zoo'
```

```
The following objects are masked from 'package:base':

    as.Date, as.Date.numeric
```

```r
mod_zip <- zeroinfl(toxin_count~exposure_level|1,data=counts,dist="poisson")
summary(mod_zip)
```

```

Call:
zeroinfl(formula = toxin_count ~ exposure_level | 1, data = counts, dist = "poisson")

Pearson residuals:
    Min      1Q  Median      3Q     Max 
-0.7012 -0.6529 -0.6135  0.1750  7.8974 

Count model coefficients (poisson with log link):
               Estimate Std. Error z value Pr(>|z|)  
(Intercept)    0.406558   0.219302   1.854   0.0638 .
exposure_level 0.007978   0.003411   2.339   0.0193 *

Zero-inflation model coefficients (binomial with logit link):
            Estimate Std. Error z value Pr(>|z|)
(Intercept)   0.2717     0.1941     1.4    0.162
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Number of iterations in BFGS optimization: 8
Log-likelihood: -202.3 on 3 Df
```

```
lrtest(mod_zip)
```

```
Likelihood ratio test

Model 1: toxin_count ~ exposure_level | 1
Model 2: toxin_count ~ 1
  #Df  LogLik Df  Chisq Pr(>Chisq)
1   3 -202.29
2   2 -205.05 -1 5.5049    0.01896 *
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
emmeans(mod_zip,~exposure_level,mode="count",infer=T, at=list(exposure_level = c(40,50)))
```

```
 exposure_level emmean    SE  df asymp.LCL asymp.UCL z.ratio p.value
             40   2.07 0.236 Inf       1.6      2.53   8.771 <0.0001
             50   2.24 0.224 Inf       1.8      2.68   9.980 <0.0001

Confidence level used: 0.95
```

```
# because the zero-inflated probability does not have any predictors, we need to compute the

coef(mod_zip)[3] |> plogis() #inverse logit transformation
```

```
zero_(Intercept)
       0.5675204
```

```
confint(mod_zip)[3,] |> plogis()
```

```
    2.5 %    97.5 %
0.4728355 0.6575168
```

```
library(pscl)
library(lmtest)

# Zero-Inflated Negative Binomial
mod_zinb <- zeroinfl(toxin_count~exposure_level|1,data=counts,dist="negbin")
summary(mod_zinb)
```

```
Call:
zeroinfl(formula = toxin_count ~ exposure_level | 1, data = counts, dist = "negbin")

Pearson residuals:
    Min      1Q  Median      3Q     Max
-0.5892 -0.5539 -0.5219  0.1675  6.5786

Count model coefficients (negbin with log link):
                Estimate Std. Error z value Pr(>|z|)
(Intercept)    -0.190448   0.490962  -0.388    0.698
exposure_level  0.010235   0.005311   1.927    0.054 .
Log(theta)     -0.251239   0.852930  -0.295    0.768

Zero-inflation model coefficients (binomial with logit link):
            Estimate Std. Error z value Pr(>|z|)
(Intercept)  -0.8807     1.2736  -0.691    0.489
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Theta = 0.7778
Number of iterations in BFGS optimization: 19
Log-likelihood: -192.4 on 4 Df
```

```
lrtest(mod_zinb)
```

```
Likelihood ratio test

Model 1: toxin_count ~ exposure_level | 1
Model 2: toxin_count ~ 1
  #Df  LogLik Df  Chisq Pr(>Chisq)
1   4 -192.43
2   3 -194.27 -1 3.6936    0.05462 .
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
emmeans(mod_zinb,~exposure_level,mode="count",infer=T, at=list(exposure_level = c(40,50)))
```

```
 exposure_level emmean    SE  df asymp.LCL asymp.UCL z.ratio p.value
             40   1.24 0.499 Inf     0.267      2.22   2.496  0.0126
             50   1.38 0.542 Inf     0.317      2.44   2.544  0.0109
```

```
Confidence level used: 0.95
```

```
# because the zero-inflated probability does not have any predictors, we need to compute the

coef(mod_zinb)[3] |> plogis() #inverse logit transformation
```

```
zero_(Intercept)
       0.2930427
```

```
confint(mod_zinb)[3,] |> plogis()
```

```
     2.5 %     97.5 %
0.03302496 0.83418815
```

1.34.3 SAS Code

```
proc genmod data=counts;
 model toxin_count =  exposure_level / dist=zip;
 zeromodel /link=logit;
 estimate ' ' @zero int 1;
run;
```

1.34.4 Answer

The analysis of the zero-inflated shows that there is weak evidence of zero-inflation in the data. There is evidence of association between toxin count and exposure level (χ^2_1=5.50,p=0.02).

1.35 Example: Hurdle Models (R)

The data set `bioChemists` contains a sample of 915 biochemistry graduate students. Suppose we want to test whether there is an association between the count of articles produced during the last 3 years of Ph.D. (`art`) and the gender (`fem`) and marital status (`mar`) of the student. Is there zero-inflation?

1.35.1 R Data Set

```
library(pscl)
data("bioChemists")
glimpse(bioChemists)
```

```
Rows: 915
Columns: 6
$ art  <int> 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0,~
$ fem  <fct> Men, Women, Women, Men, Women, Women, Women, Men, Men, Women, Men~
$ mar  <fct> Married, Single, Single, Married, Single, Married, Single, Marrie~
$ kid5 <dbl> 0, 0, 0, 1, 0, 2, 0, 2, 0, 0, 0, 0, 1, 0, 0, 0, 0, 2, 0, 0, 0, 3,~
$ phd  <dbl> 2.520, 2.050, 3.750, 1.180, 3.750, 3.590, 3.190, 2.960, 4.620, 1.~
$ ment <int> 7, 6, 6, 3, 26, 2, 3, 4, 6, 0, 14, 13, 3, 4, 0, 1, 7, 13, 7, 9, 6~
```

1.35.2 R Code

```
mod_poisson <- glm(art~fem+mar,data=bioChemists,family="poisson")
library(performance)
check_zeroinflation(mod_poisson)
```

```
# Check for zero-inflation

   Observed zeros: 275
  Predicted zeros: 172
            Ratio: 0.63

Model is underfitting zeros (probable zero-inflation).
```

The difference in observed and predicted zeros is large, hence there might be zero-inflation. Note that the zero counts indicate that the student has published a manuscript. Once a student publishes a manuscript, it is impossible for them to record a zero count for articles published. This is why we need a hurdle model.

```
mod_hurdle <- hurdle(art~fem+mar|fem+mar,data=bioChemists,dist="poisson")
summary(mod_hurdle)
```

```
Call:
hurdle(formula = art ~ fem + mar | fem + mar, data = bioChemists, dist = "poisson")
```

```
Pearson residuals:
    Min      1Q  Median      3Q     Max
-1.1392 -1.0178 -0.3446  0.3905 10.2843

Count model coefficients (truncated poisson with log link):
             Estimate Std. Error z value Pr(>|z|)
(Intercept)  0.847598   0.064084  13.226  < 2e-16 ***
femWomen    -0.237351   0.064199  -3.697 0.000218 ***
marMarried   0.008846   0.066944   0.132 0.894867
Zero hurdle model coefficients (binomial with logit link):
            Estimate Std. Error z value Pr(>|z|)
(Intercept)  0.90794    0.15640   5.805 6.42e-09 ***
femWomen    -0.24195    0.14923  -1.621    0.105
marMarried   0.07802    0.15636   0.499    0.618
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Number of iterations in BFGS optimization: 9
Log-likelihood: -1670 on 6 Df
```

```
#let's test each variable
mod_fem <- update(mod_hurdle,.~fem)# gender only model
mod_mar <- update(mod_hurdle,.~mar) # marital status only model
lrtest(mod_hurdle,mod_fem)
```

```
Likelihood ratio test

Model 1: art ~ fem + mar | fem + mar
Model 2: art ~ fem
  #Df  LogLik Df  Chisq Pr(>Chisq)
1   6 -1670.1
2   4 -1670.2 -2 0.2659     0.8755
```

```
lrtest(mod_hurdle,mod_mar)
```

```
Likelihood ratio test

Model 1: art ~ fem + mar | fem + mar
Model 2: art ~ mar
  #Df  LogLik Df  Chisq Pr(>Chisq)
```

```
1   6 -1670.1
2   4 -1678.4 -2 16.602  0.0002483 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

1.35.3 Answer

There is strong evidence of association between the count of published articles in the final 3 years of Ph.D. and marital status ($\chi^2_2 = 16.6$, p=0.0002), but not with gender (χ^2_2=0.266, p=0.8755).